

<https://presidencyuniversity.in/facility/library/>

Topics relevant to “EMPLOYABILITY SKILLS”:

Handling Unstructured Data for Employability skills through Experiential Learning techniques. This is attained through the assessment component mentioned in the course handout.

Course Code: CSE 3035	Course Title: R Programming For Data Science Type of Course: Integrated	L- T-P- C	1 -0	4	3
Version No.	1				
Course Pre-requisites	NIL				
Anti-requisites	NIL				
Course Description	This course is designed to provide the core concepts of data analytics in the R environment. Initially train them with basic R, then progressively increase the difficulty as they move along in the course, capping with advanced techniques through case studies. Mastering the core concepts and techniques of data analytics in R, will help the students to apply their knowledge to a wide range of Data Analytics. R is now considered one of the most popular analytics tools in the world.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of R Programming For Data Science and attain Skill Development through Experiential Learning techniques.				
Course Out Comes	On successful completion of this course the students shall be able to: Apply basic R functions pertaining to fundamental data analysis. [Application] Interpret data using appropriate statistical methods [Application] Demonstrate the decision trees concept with the given dataset. [Application] Demonstrate the Mining concepts for both Data and Text. [Application]				
Course Content:					
Module 1	Introduction	Assignment	Data Collection/Interpretation	6 Sessions	
Topics: Introduction to R, Overview of data analysis, Working with directory in R, Loading and handling data in R, Data Visualization with ggplot2, Data Transformation with dplyr.					
Module 2	Exploratory Data Analysis	Coding Assignment	Case Study	11 Sessions	

Topics: Exploring a new dataset, Anomalies in numerical data, Visualizing relations between variables, Assumptions of Linear Regression, Validating Linear Assumption, Missing Values, Covariation, Patterns and Models, ggplot2 Calls.				
Module 3	Regression Analysis	Coding Assignment	Project	12 Sessions
Topics: Introduction, Types of Regression Analysis Models, Linear Regression, Simple Linear Regression, Non-Linear Regression, Regression Analysis with Multiple Variables, Cross Validation, Principal Component Analysis, Factor Analysis.				
Module 4	Classification	Quiz	Project	8 Sessions
Topics: Introduction, Different types of Classification, Logistic Regression, Support Vector Machines, K-Nearest Neighbors, Naïve Bayes Classifier, Decision Tree Classification, Random Forest Classification, Evaluation.				
List of Laboratory Tasks: 1. Using with and without R objects on console 2. Using mathematical functions on console 3. Write an R script, to create R objects for calculator 4. Write an R script to find basic descriptive statistics using summary, str, quartile function on mtcars& cars datasets. 5. Reading different types of data sets (.txt, .csv) from Web and disk and writing in file in specific disk location. b. Reading Excel data sheet in R 6. Find the data distributions using box and scatter plot. 7. Find the outliers using plot. 8. Plot the histogram, bar chart and pie chart on sample data 9. Find the correlation matrix. 10. Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data 11. Create a regression model for a given dataset 12. Install relevant package for classification. 13. Choose classifier for classification problem. c. Evaluate the performance of classifier. 14. Install relevant package for classification. 15. Choose classifier for classification problem. c. Evaluate the performance of classifier.				
Targeted Application & Tools that can be used Tools: RStudio / Google Colab				
Project work/Assignment:				

Assignment:

During the course, students would need to do coding assignments to learn to train and use different models. Sample coding assignments include:

Analysis of Sales Report of a Clothes Manufacturing Outlet.

Comcast Telecom Consumer Complaints.

Web Data Analysis

Text Book

T1 Hadley Wickham and Garrett Grolemund, "R for Data Science", O'reilly, 2017.

References

R1 Dr. Bharati Motwani, "Data Analytics using R", Wiley, 2019.

Web resources:

<https://www.geeksforgeeks.org/r-programming-for-data-science/>

<https://r4ds.had.co.nz/>

Topics relevant to "SKILL DEVELOPMENT": Regression model, classifier for Skill Development through Experiential Learning techniques. This is attained through assessment component mentioned in course handout.